Supplementary Material

Effect of micro-aeration on syntrophic and methanogenic activity in anaerobic sludge

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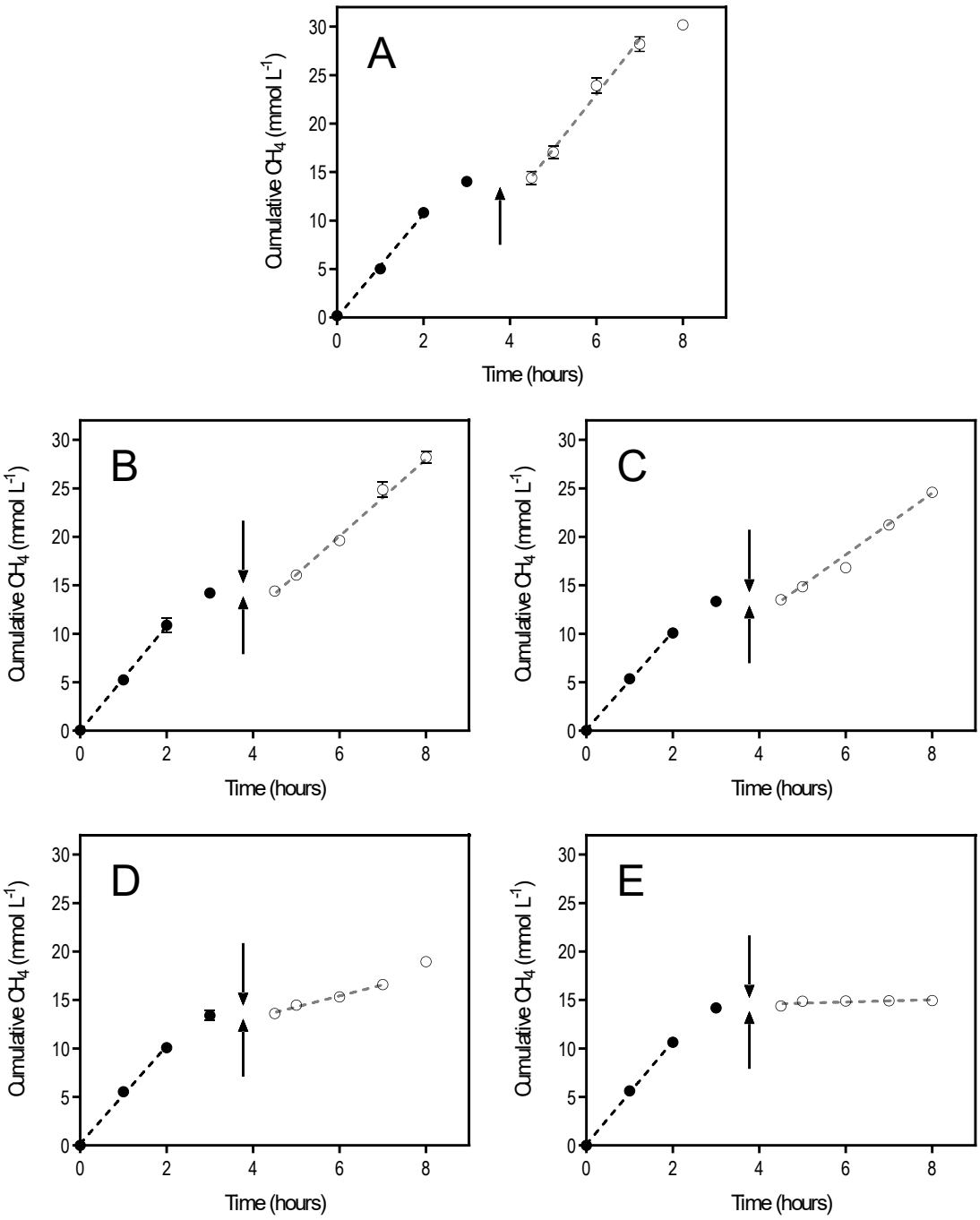

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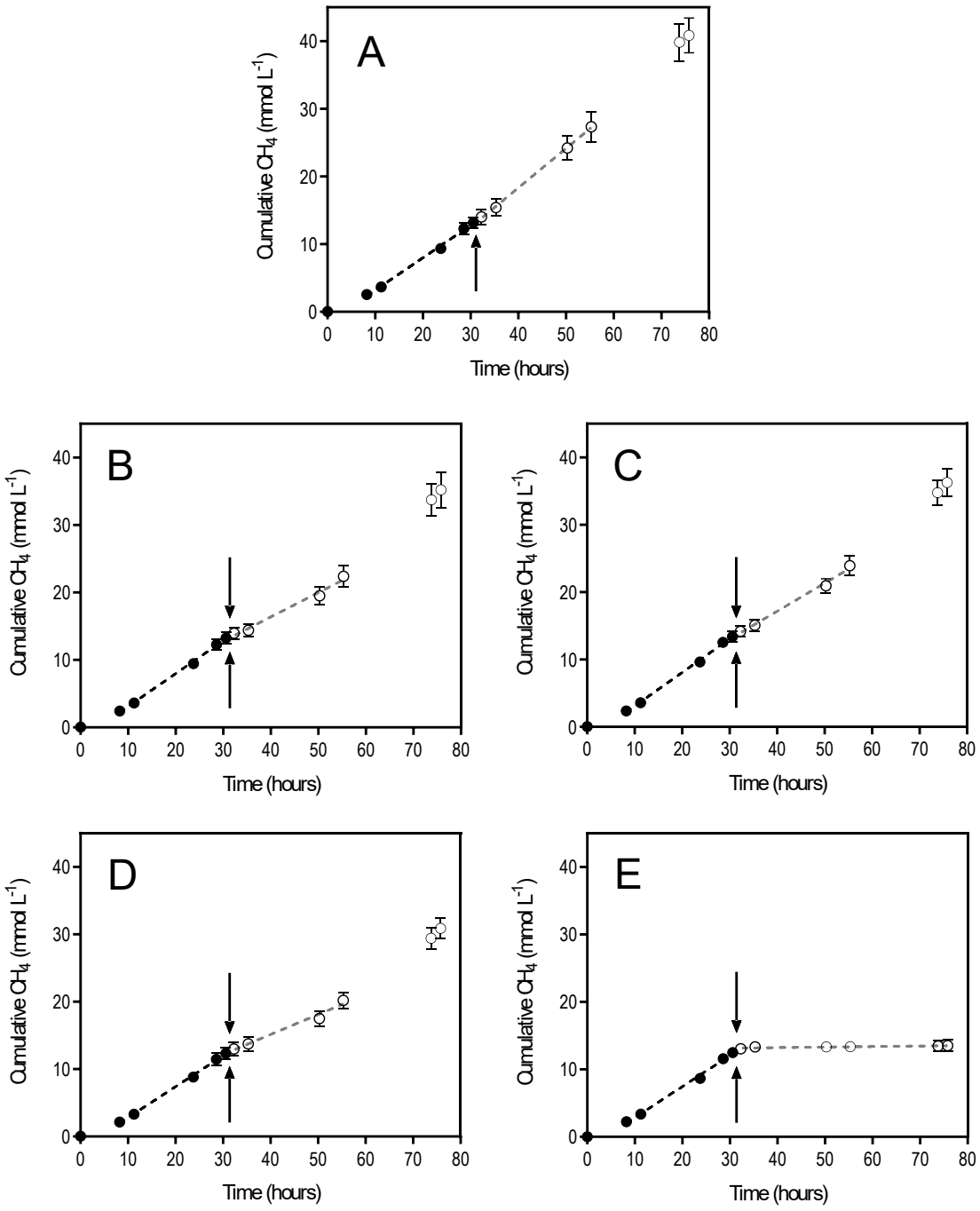

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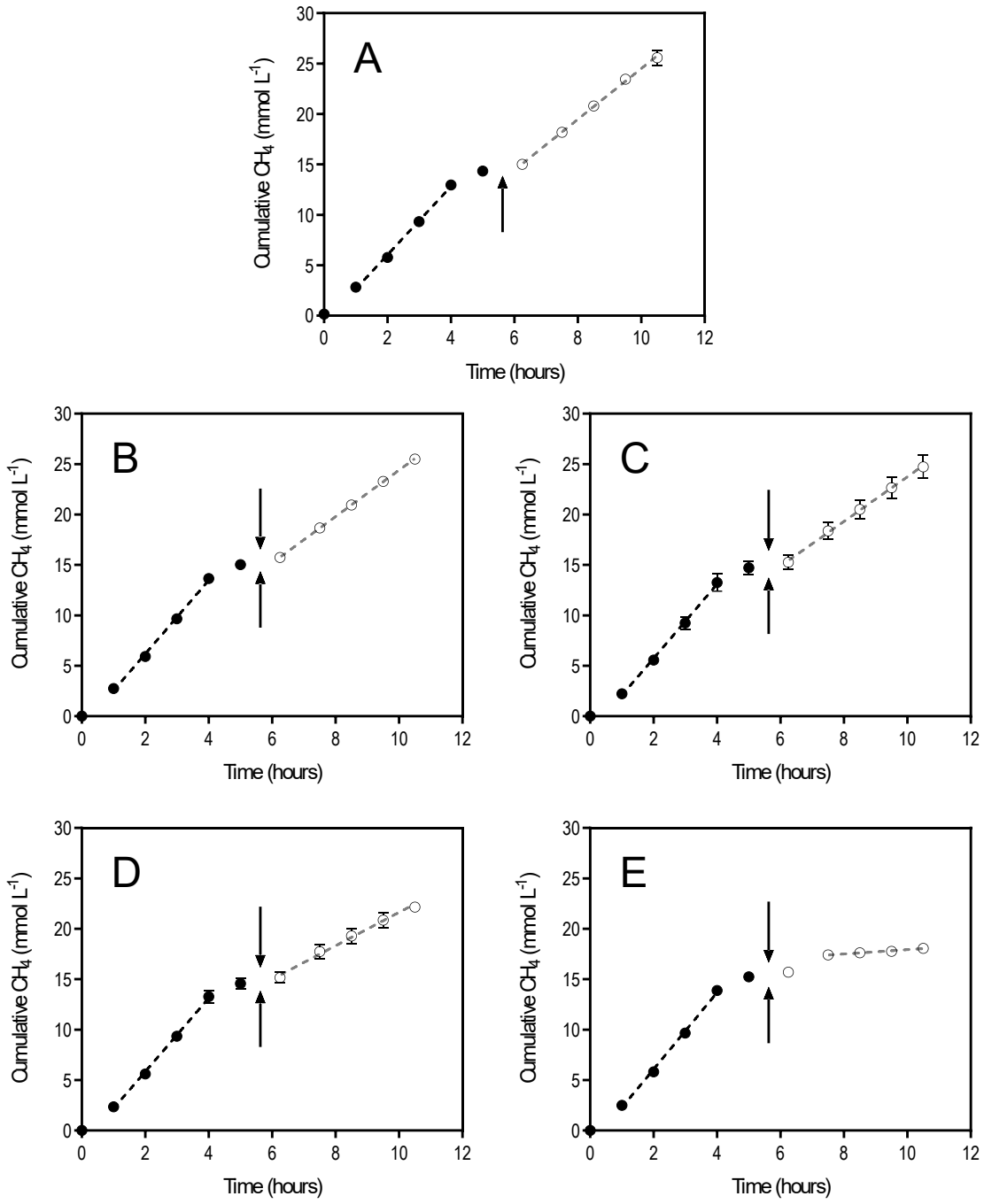

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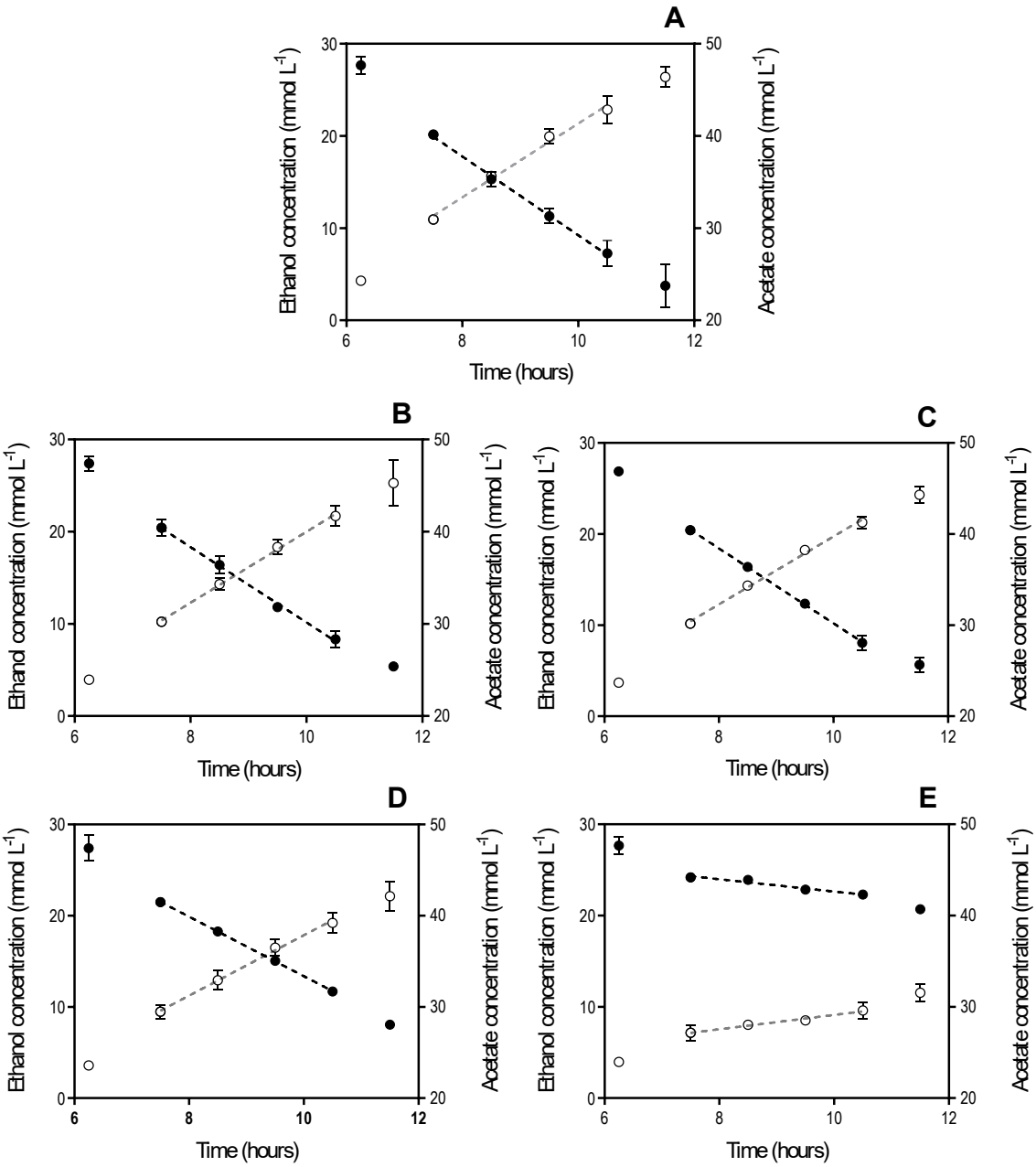

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Table S1. H_2 and acetate concentrations at the beginning of P1 and P2, and total consumption (average of triplicates ± standard deviation).

Substrate	P1		P2		
	Initial concentration (mmol L^{-1}) [(a)]	Total consumption (mmol L^{-1})	O_2 (%)	Initial concentration (mmol L^{-1}) [(a)]	Total consumption (mmol L^{-1})
H_2/CO_2	55.6 ± 0.6	48.6 ± 2.2	0	53.2 ± 1.2	49.3 ± 1.4
	53.5 ± 0.5	48.2 ± 1.5	0.5	51.6 ± 0.5	43.5 ± 1.5
	53.4 ± 0.3	44.9 ± 0.7	1.0	48.5 ± 2.3	32.8 ± 2.6
	53.2 ± 0.3	46.0 ± 1.1	2.5	46.1 ± 0.6	12.9 ± 1.1
	53.5 ± 0.5	48.3 ± 3.6	5.0	42.2 ± 0.2	0.0 ± 0.0
Acetate	24.3 ± 0.6	12.1 ± 1.5	0	25.4 ± 2.1	24.0 ± 2.6
	25.2 ± 0.3	12.6 ± 0.7	0.5	25.9 ± 0.5	17.6 ± 1.5
	23.7 ± 2.0	11.4 ± 2.0	1.0	25.2 ± 0.6	19.3 ± 0.7
	24.0 ± 0.8	10.6 ± 1.2	1.5	25.9 ± 0.7	16.0 ± 1.3
	24.7 ± 0.9	11.7 ± 1.3	2.0	26.6 ± 1.1	0.4 ± 1.9

[(a)] H_2 concentration was measured in H_2/CO_2 assays.

Table S2. Ethanol and acetate concentrations measured at the beginning (t_0) and end (t_f) of P1 and P2, at increasing O_2 concentrations, in the assays with ethanol. Values are the average of triplicates ± standard deviation.

P1				P2				
[Eth] t_0 (mmol L^{-1})	**[Eth] t_f (mmol L^{-1})**	**[Ac] t_0 (mmol L^{-1})**	**[Ac] t_f (mmol L^{-1})**	**O_2 (%)**	**[Eth] t_0 (mmol L^{-1})**	**[Eth] t_f (mmol L^{-1})**	**[Ac] t_0 (mmol L^{-1})**	**[Ac] t_f (mmol L^{-1})**
26.6 ± 0.0	0.0 ± 0.0	0.1 ± 0.0	24.9 ± 0.2	0	27.7 ± 0.8	3.8 ± 1.9	24.3 ± 0.3	46.4 ± 0.9
26.1 ± 0.4	0.0 ± 0.0	0.1 ± 0.0	24.6 ± 0.4	0.5	27.4 ± 0.7	5.4 ± 0.4	23.9 ± 0.6	45.3 ± 2.0
26.1 ± 0.4	0.0 ± 0.0	0.1 ± 0.0	24.4 ± 0.1	1.0	26.9 ± 0.4	5.6 ± 0.7	23.7 ± 0.1	44.3 ± 0.7
25.7 ± 0.6	0.0 ± 0.0	0.1 ± 0.0	24.2 ± 0.5	2.5	27.4 ± 1.1	7.2 ± 1.2	23.6 ± 0.6	42.1 ± 1.1
26.2 ± 0.3	0.0 ± 0.0	0.1 ± 0.0	24.3 ± 0.2	5.0	27.7 ± 0.8	20.7 ± 0.4	24.0 ± 0.1	31.6 ± 0.8

Eth (Ethanol), Ac (Acetate)